Introduction

In this laboratory experiment, we will learn circuit analysis and diode fundamentals. Our objective is to exploring the behavior of electrical circuits and the specific characteristics of diodes. We transition from theoretical simulations to experiments.

1- Prelab

1.1- Preliminary work (Circuit Analysis)

In the preliminary work, we are supposed to create the following circuit on the program LTSpice. Then investigate the behaviors of the circuit components, in terms of Voltage (V) and Current (I).

1. First, we have correctly set the circuit on LTSpice and run the DC analysis with. op command. The circuit has 4, 1k resistor and 2 voltage sources.

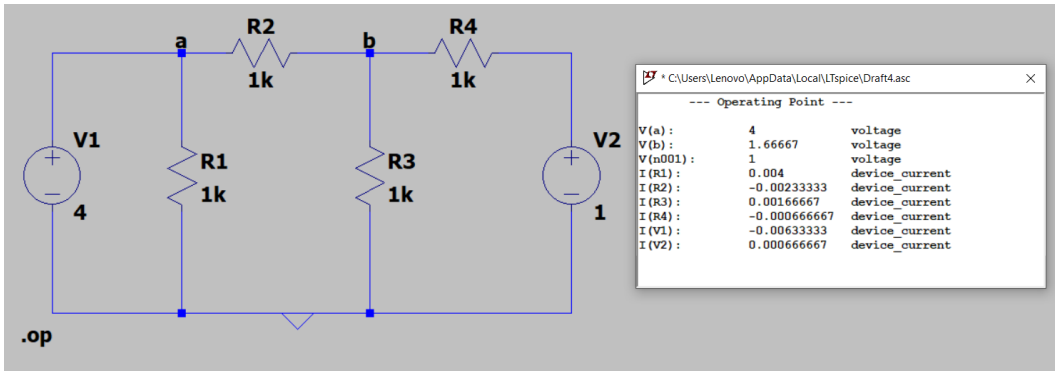


Figure 1 Task 1.1 Circuit 1.1 on the LTSpice program with results.

As we can see from figure 1. .op command shows us the Operating Point with values of points such as point (a) and point (b). These points have their own voltage on those points. Point (a) has 4 volts and point (b) has 1.7 volt. Also every resistor has a current passes through them.

R1	0.004 Amp
R2	-0.0023 Amp
R3	0.0017 Amp
R4	-0.00067 Amp
V1	-0.0063 Amp
V2	0.00067 Amp

1.2- Transient analysis

For the transient analysis, there is usually time varying sources. So, we have changed DC voltage sources to AC voltage sources. More specifically sinusoidal voltage sources. We simply just replaced DC voltage sources with AC.

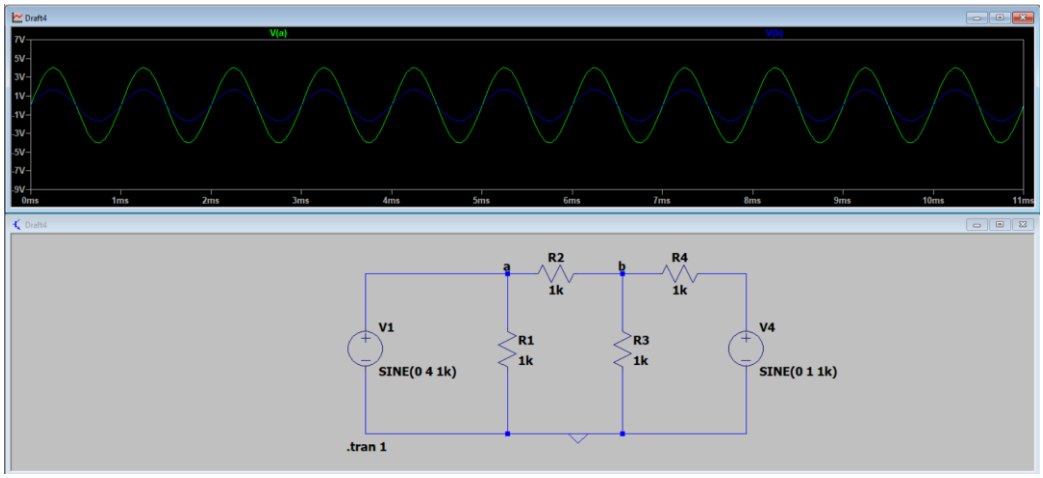


Figure 2 Transient Analysis of the circuit 1.1 Showing V(a) and V(b) graph.

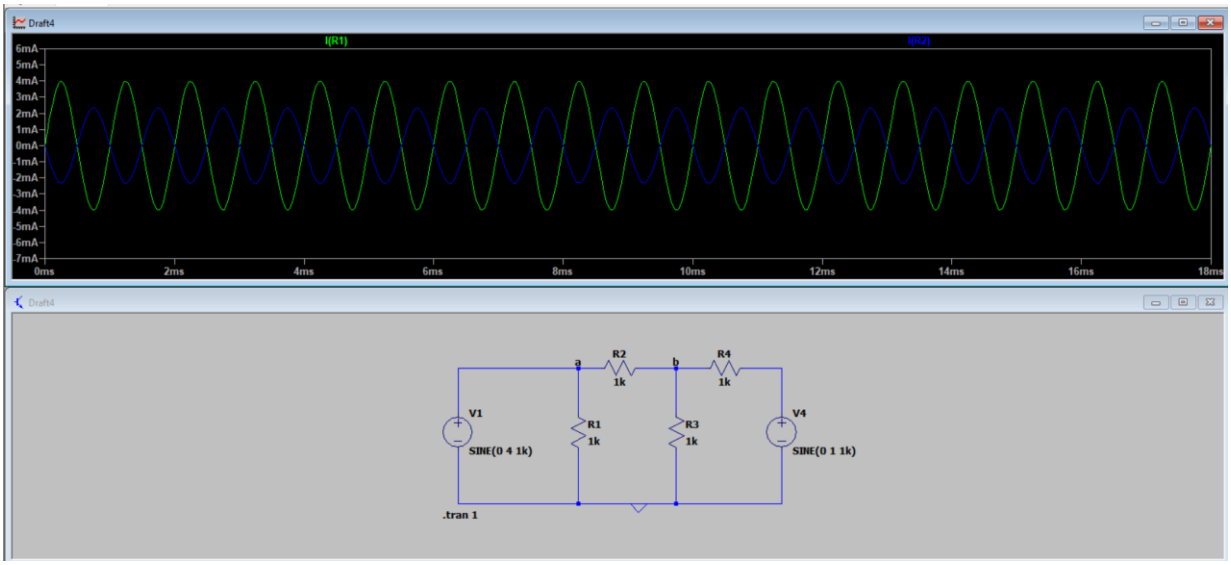


Figure 3 Transient Analysis of the circuit 1.1 Showing I(R1) and I(R2) graph.

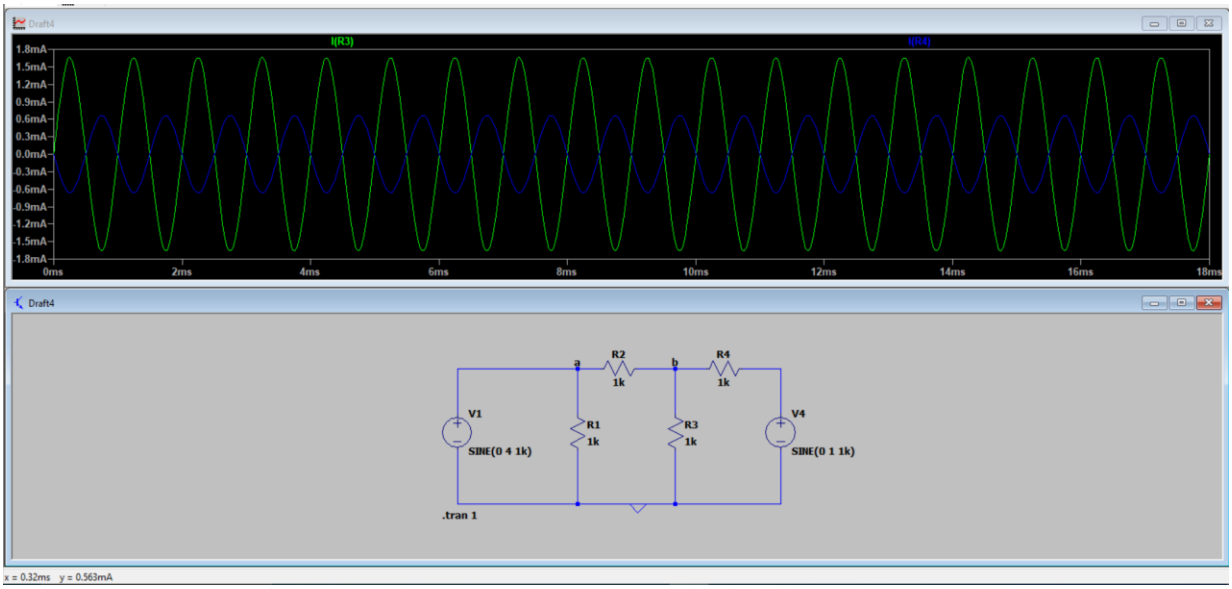


Figure 4 Transient Analysis of the circuit 1.1 Showing I(R3) and I(R4) graph.

2- Preliminary work (Diode Basics)

2.1 **I-V Curve of a diode:** In the preliminary work we have constructed a simple diode circuit (Circuit 2.1) with 1k Ohm resistance with 5V voltage source completed with 1N4148 diode and ploted I-V curves using spice.

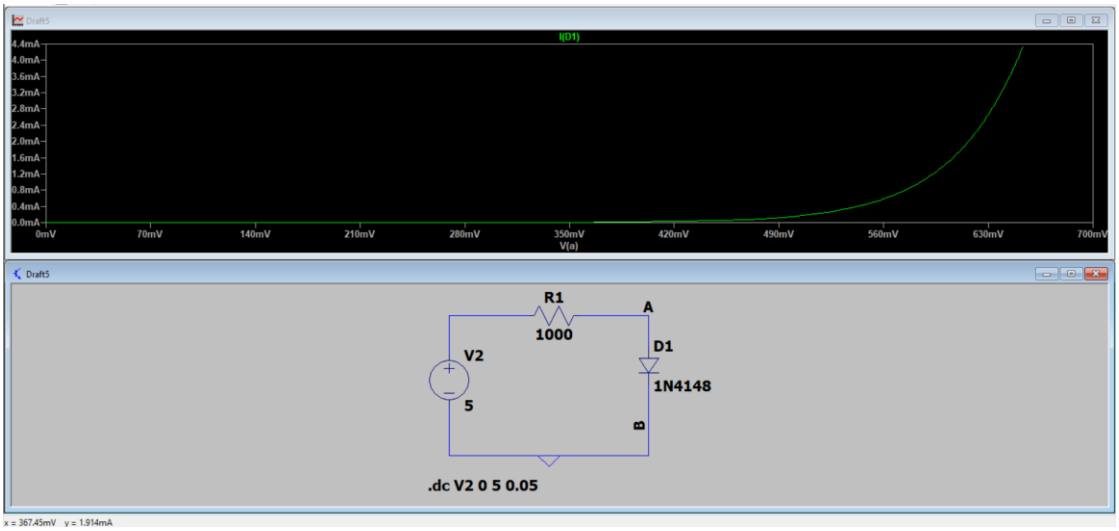


Figure 5

3- Lab Tasks

3.1. DC measurements

1. Firstly we have constructed circuit 2.1. from the lab manual.

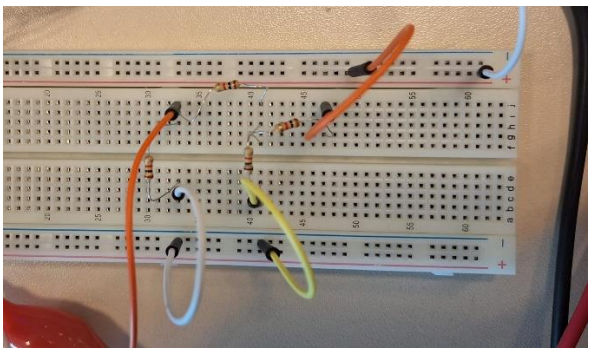


Figure 6 Circuit 2.1 on breadboard.

2. When we measure the same points on LTSpice and on breadboard, we get that table below the pictures of measurements.

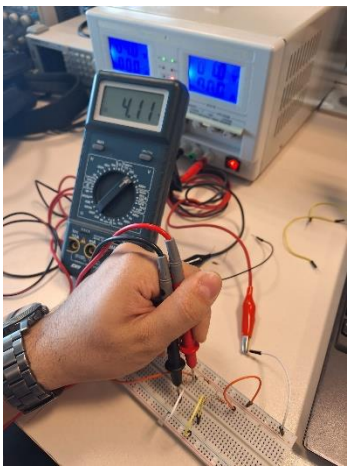


Figure 7 Measurements of node A

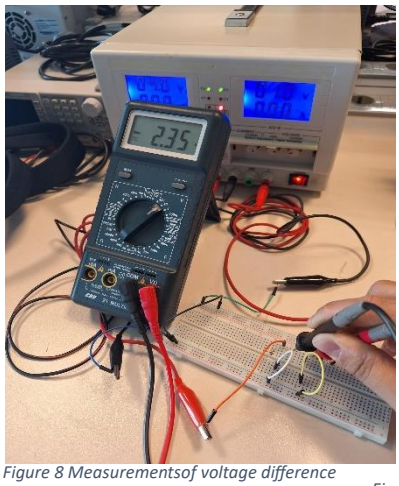


Figure 8 Measurementsof voltage difference on the R2.

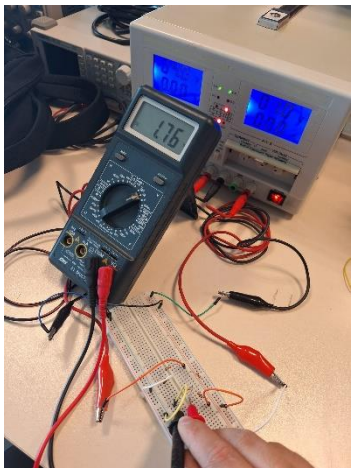


Figure 9 Measurements of node B

	On The LTSpice	In the lab measurements
A	4V	4.11V
B	1.67V	1.76V
I(R2)	-2.34mA	-2.35mA

We have measured the voltage differences between two side of the resistor and we calculated the current passes through the resistor.

I=V/R is our formula to find current. So we have got (-2.35V) on the multimeter that means I=(-2.35)/1000 and that gives -2.35 mAmp for the current passes through the resistor (R2).

There is some differences between LTSpice results and Lab measurements, but that is a negligible difference. They are happening because of some factors such as temperature, handshakes while measurement etc.

3.2 I-V curve

1. Firstly we have constructed circuit 3.1. from the lab manual. That circuit has 1V input and 10k resistor only.

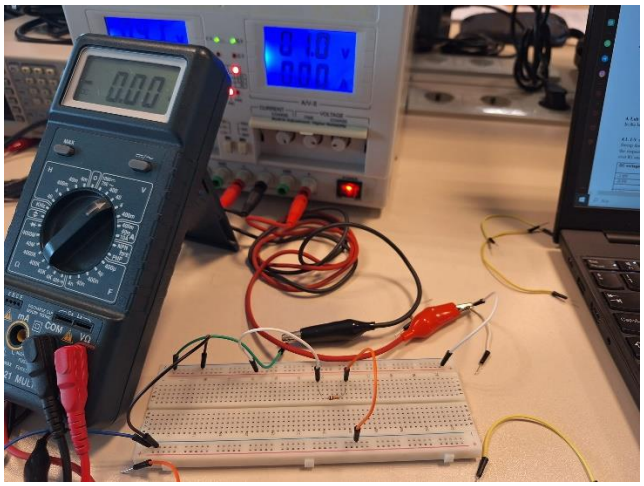
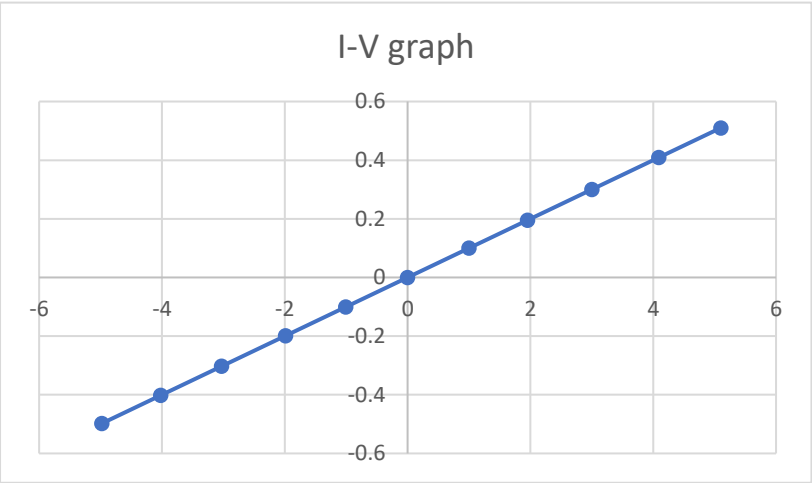


Figure 10 Circuit 3.1 on the breadboard.

3. In that task we analyze the electrical properties of a 10k ohm resistor when subjected to varying voltage levels within the range of 5V to -5V, incrementally adjusted in 1V intervals. Then we measure Voltage differences on each side of 10k resistor.

   All of our measurements for the circuit 3 is on the table, shown below.

V	Measured V with multimeter (V)	I=V/R (mAmp) (R=10k)
-5	-4.98	-0.498
-4	-4.02	-0.402
-3	-3.03	-0.303
-2	-1.99	-0.199
-1	-1.01	-0.101
0	0.00	0.00
1	1.00	0.100
2	1.95	0.195
3	3.00	0.300
4	4.09	0.409
5	5.10	0.510



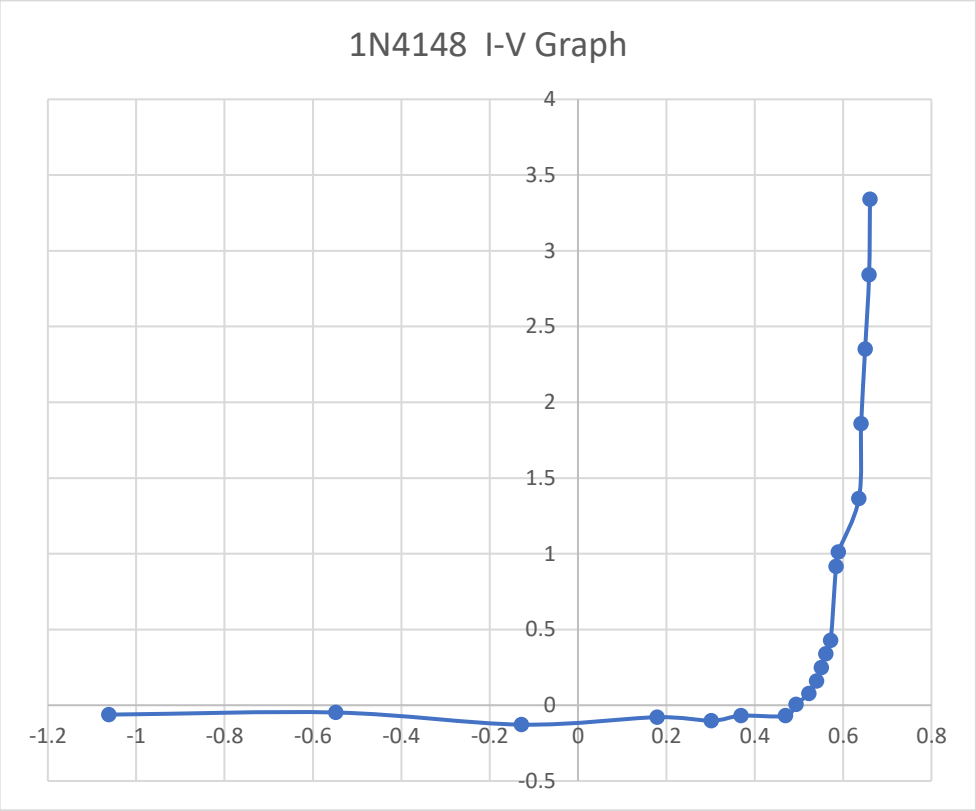
4. We can observe that Current and Voltage values are directly proportional to each other. Also, there is not any curve for the resistor on the I-V graph.

4- Lab Tasks

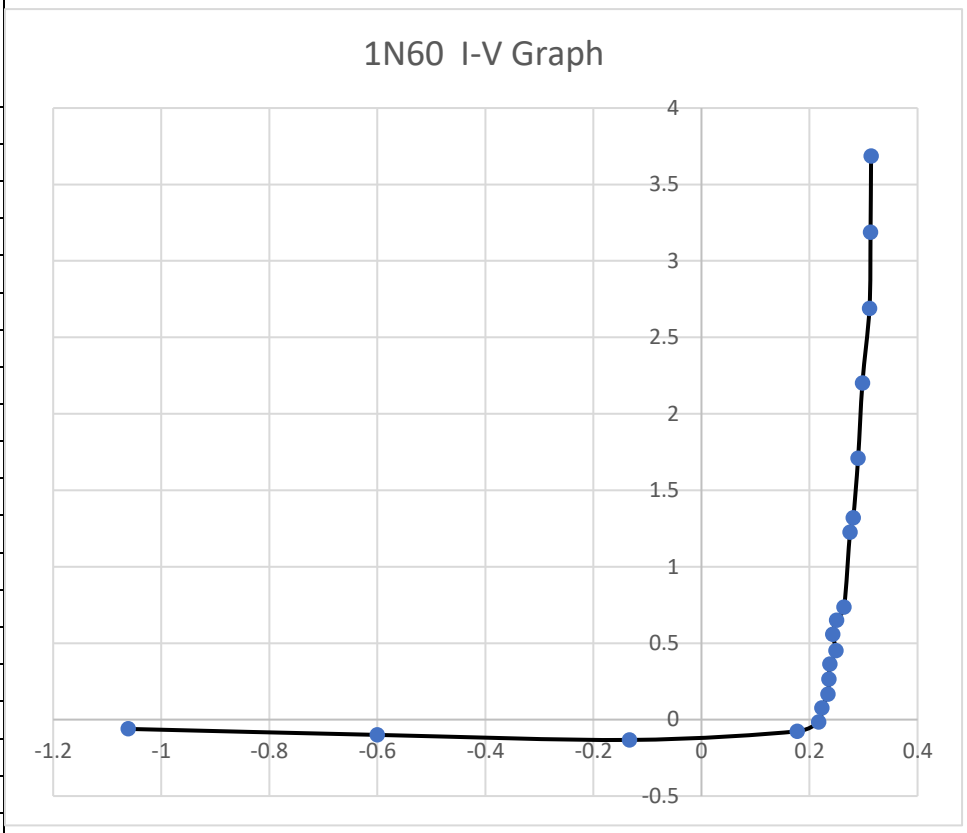
4.1. I-V curves

1. For the task 4, we will look at the how different diodes behave differently. We have tried diodes 1N4148, 1N60, Zener diodes and LED to observe their behaviors as a diode. To analyze the behaviors we have built the circuit 2.1 as shown in the lab manual.
2. Next, we measured the Voltage difference on the Resistor with the multimeter, then we have changed voltage between -1V to 4V (Except Zener, we have used -10V to 0V because Zener diode creates extra reverse breakdown voltage.)

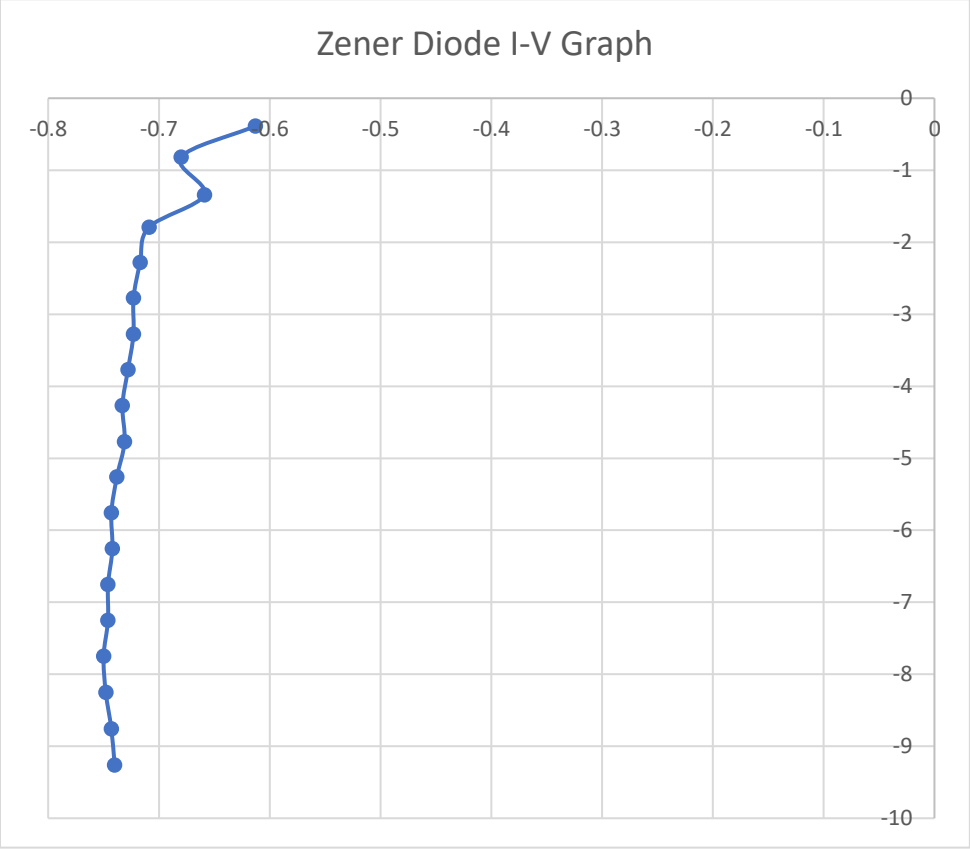
1N4148			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-1	0.00	-1.06	-0.06
-0.5	0.00	-0.55	-0.05
0	0.00	-0.13	-0.13
0.1	0.00	0.18	-0.08
0.2	0.00	0.30	-0.10
0.3	0.00	0.37	-0.07
0.4	0.05	0.47	-0.07
0.5	0.10	0.49	0.01
0.6	0.17	0.52	0.08
0.7	0.28	0.55	0.15
0.8	0.33	0.55	0.25
0.9	0.43	0.56	0.34
1	0.51	0.57	0.43
1.5	1.03	0.58	0.92
1.6	1.04	0.59	1.01
2	1.45	0.64	1.36
2.5	1.98	0.64	1.86
3	2.40	0.65	2.35
3.5	2.48	0.66	2.84
4	3.28	0.66	3.34



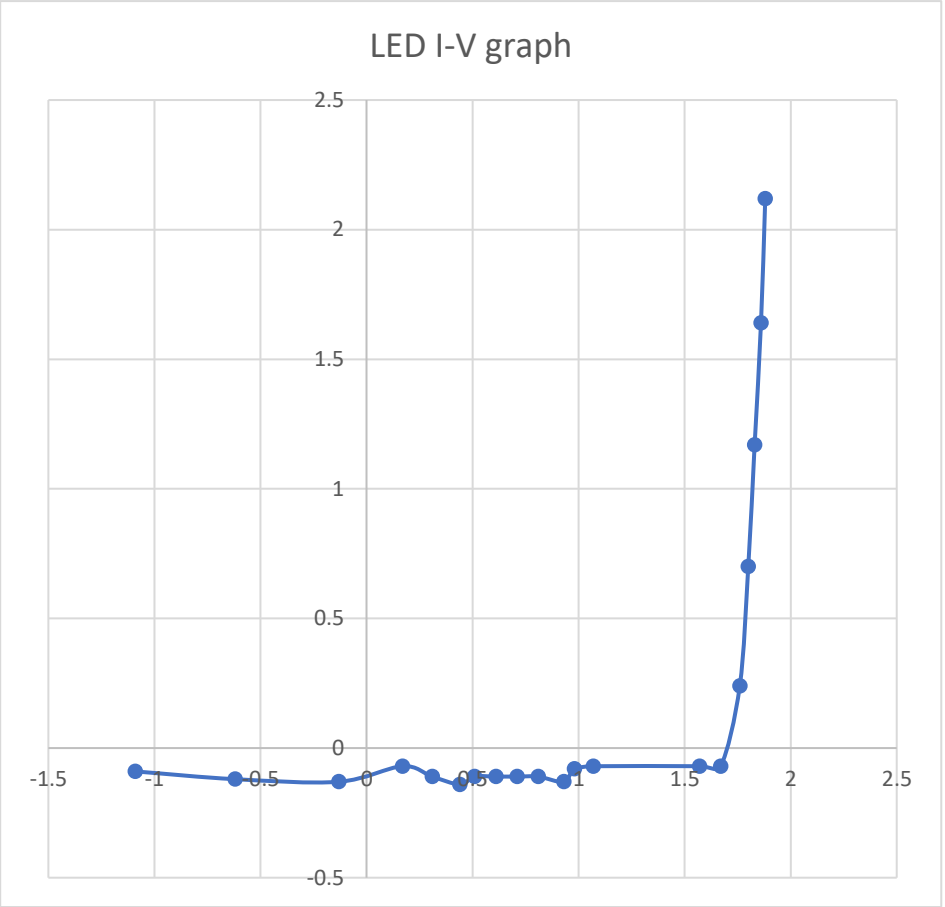
1N60			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-1	0.00	-1.06	-0.06
-0.5	0.00	-0.60	-0.10
0	0.00	-0.13	-0.13
0.1	0.00	0.18	-0.08
0.2	0.09	0.22	-0.02
0.3	0.11	0.23	0.08
0.4	0.15	0.23	0.17
0.5	0.20	0.23	0.27
0.6	0.30	0.24	0.37
0.7	0.33	0.25	0.45
0.8	0.57	0.24	0.56
0.9	0.65	0.25	0.65
1	0.76	0.26	0.74
1.5	1.22	0.28	1.22
1.6	1.36	0.28	1.32
2	1.68	0.29	1.71
2.5	2.17	0.30	2.21
3	2.68	0.31	2.69
3.5	2.92	0.32	3.18
4	3.55	0.31	3.69



ZENER			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-10	-8.00	-0.74	-9.26
-9.5	-8.01	-0.74	-8.76
-9	-6.67	-0.74	-8.26
-8.5	-7.01	-0.75	-7.75
-8	-6.69	-0.75	-7.26
-7.5	-6.56	-0.75	-6.75
-7	-6.05	-0.74	-6.26
-6.5	-5.54	-0.74	-5.76
-6	-4.89	-0.73	-5.27
-5.5	-3.73	-0.73	-4.77
-5	-3.93	-0.73	-4.27
-4.5	-3.33	-0.72	-3.78
-4	-2.99	-0.72	-3.28
-3.5	-2.79	-0.72	-2.78
-3	-2.08	-0.72	-2.28
-2.5	-1.74	-0.72	-1.78
-2	-0.74	-0.66	-1.34
-1.5	-0.83	-0.68	-0.82
-1	-0.33	-0.61	-0.39



LED			
DC Voltage (V)	R1 Voltage (V)	D1 Voltage (V)	D1 Current (mA)
-1	0.00	-1.09	-0.09
-0.5	0.00	-0.62	-0.12
0	0.00	-0.13	-0.13
0.1	0.00	0.17	-0.07
0.2	0.00	0.31	-0.11
0.3	0.00	0.44	-0.14
0.4	0.00	0.51	-0.11
0.5	0.00	0.61	-0.11
0.6	0.00	0.71	-0.11
0.7	0.00	0.81	-0.11
0.8	0.00	0.93	-0.13
0.9	0.00	0.98	-0.08
1	0.00	1.07	-0.07
1.5	0.00	1.57	-0.07
1.6	0.05	1.67	-0.07
2	0.31	1.76	0.24
2.5	0.72	1.80	0.70
3	1.12	1.83	1.17
3.5	1.68	1.86	1.64
4	1.95	1.88	2.12



- Our observations highlight the distinctive characteristics of diodes, revealing distinct threshold voltages among them. But Zener behaves differently because of the additional reverse breakdown voltage.

TASK 4 is missing due the limited time. I was not able to finish the last task. So that I am not able to write anything about task 4.

But by the data we have gathered by 3 lab and 1 pre-lab task we can create simple conclusion.

Conclusion

In this laboratory experiment, we explored circuit analysis and diode fundamentals. We began by simulating circuits and then validating them through practical measurements. Circuit analysis involved DC and transient analyses. We observed how DC components affect node voltages and current flow, while transient analysis introduced us to time-varying sources. In the diode basics experiments, we studied I-V curves for various diodes, emphasizing threshold voltages and saturation currents. The Zener diode was thoroughly examined, revealing its reverse breakdown voltage.

By that experiment we learned

- Differentiating between various diodes, such as 1N4148, 1N60, LED, and Zener diodes, and understanding their unique characteristics.
- Understanding the principles of DC analysis, which involves calculating voltages and currents for circuits with only DC inputs.
- Interpreting I-V curves for diodes and identifying threshold voltages and saturation currents.
- Discovering the reverse breakdown voltage of a Zener diode and its distinctive behavior.